

Data Construction Pipeline

1

Source
Video
Selection



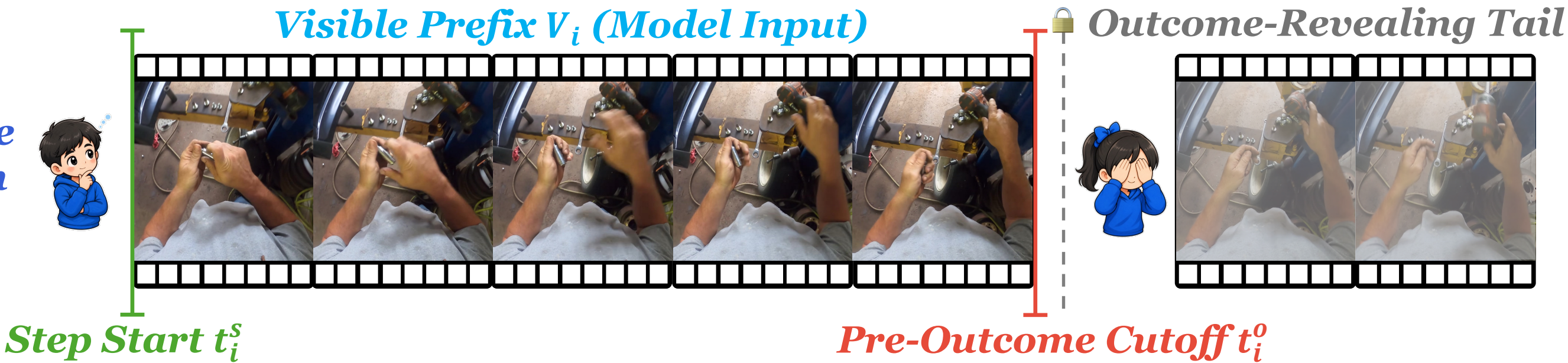
2

Manual
Micro-Step
Segmentation



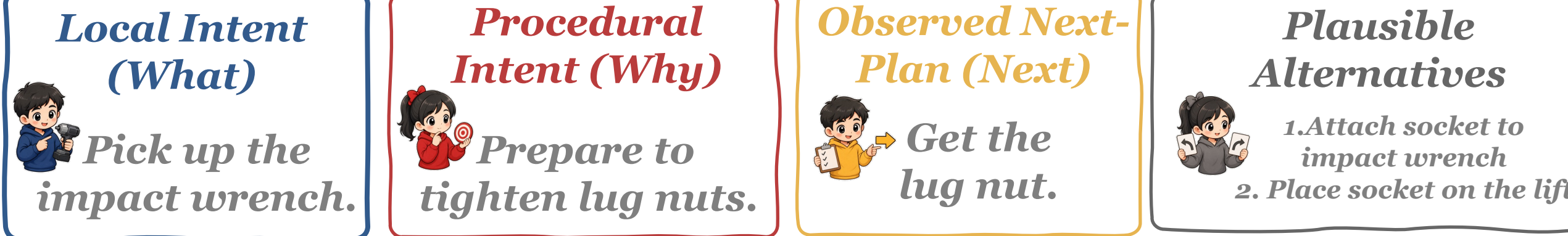
3

Pre-Outcome
Observation
Cutoff



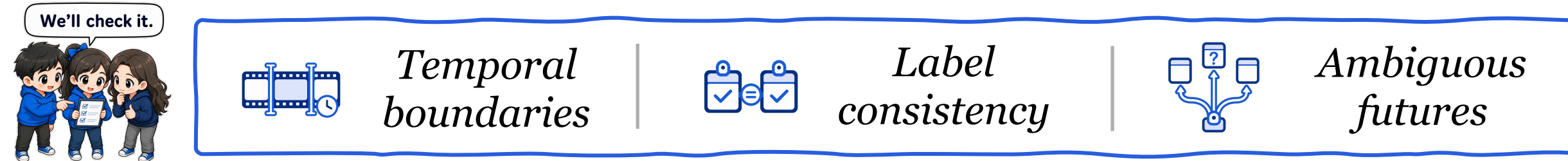
4

What- Why-
Next
Annotation

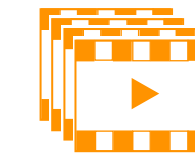


5

Multi-Round
Human Review



Benchmark Statistics



3,014
Micro-Steps



32
Source Videos



15
Scenarios



≤ 10 s
Per Step

Scenario Coverage



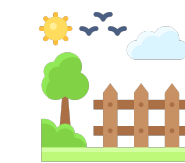
Clothing
Closet



Garden



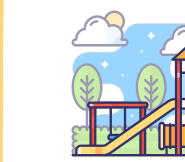
Farm



Yard



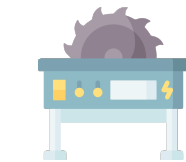
Deck



Space



Kitchen



Workshop



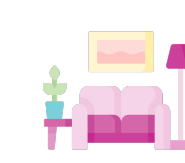
Garage



Bedroom



Study
Room



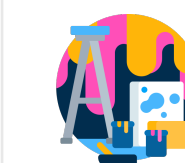
Living
Room



Laundry
Room



Hallway



Art
Studio